Lesson 4.1 - Quartiles, Deciles, and Percentiles

> **Video:** Quartiles, Deciles, & Percentiles With Cumulative Relative Frequency by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

What they are

Measure	Splits the sorted data into	Memory aid
Quartiles Q_1, Q_2, Q_3	**4** equal parts	4 quarters = 1 dollar
Deciles $D_1, \ldots, D_9$	**10** equal parts	"deci" = one tenth (10 decimeters = 1 meter)
Percentiles $P_1, \ldots, P_{99}$	**100** equal parts	per-**cent**

They line up with each other:

```
Percentile:  0   10   20   25   30   40   50   60   70   75   80   90  100
Decile:      D0  D1   D2        D3   D4   D5   D6   D7        D8   D9  D10
Quartile:    min           Q1             Q2                  Q3         max
```

- $Q_1 = P_{25}, \quad Q_2 = D_5 = P_{50} = \text{median}, \quad Q_3 = P_{75}$
- $D_k = P_{10k}$, so $D_4 = P_{40}$ and $D_9 = P_{90}$.

What does a percentile mean?

The **70th percentile** is a value such that **70%** of the data is **at or below** it and **30%** is **at or above** it.

Example: if your SAT score is at the 70th percentile, you scored as well as or better than 70% of test-takers, and 30% scored as well as or better than you.

Method 1 - quartiles as medians of halves

- Q_2 = median of the whole data set.
- Q_1 = median of the **lower half**; Q_3 = median of the **upper half**. When n is odd, the median is left out of both halves.

Example (odd $n = 11$): 2, 3, 5, 7, 8, **10**, 11, 13, 15, 16, 19

- $Q_2 = 10$
- Lower half 2, 3, **5**, 7, 8 → $Q_1 = 5$
- Upper half 11, 13, **15**, 16, 19 → $Q_3 = 15$

Method 2 - the percentile location formula

The **position** of the k-th percentile in the sorted list is

$$L_k = \frac{k}{100}(n+1)$$

- If L_k is a whole number, the percentile is the value at that position.
- If L_k falls between two positions, the video **averages the two neighboring values**.

Same data ($n = 11$):

Percentile	Location	Value
$P_{25} = Q_1$	$\frac{25}{100}(12) = 3$ → 3rd value	5
$P_{50} = Q_2$	$\frac{50}{100}(12) = 6$ → 6th value	10
$P_{75} = Q_3$	$\frac{75}{100}(12) = 9$ → 9th value	15

Both methods agree.

Example with even $n = 16$

The video uses a list of 16 numbers shown on screen. The positions that matter are:

Position	4	5	6	8	9	10	11	12	13
Value	9	11	12	15	16	16	17	19	20

- **Medians of halves:** $Q_2 = \dfrac{15+16}{2} = 15.5,\ Q_1 = \dfrac{9+11}{2} = 10,\ Q_3 = \dfrac{19+20}{2} = 19.5$
- **Formula for Q_1:** $L_{25} = \frac{1}{4}(17) = 4.25$, between the 4th and 5th values → $\dfrac{9+11}{2} = 10$ ✓
- **Formula for Q_3:** $L_{75} = \frac{3}{4}(17) = 12.75$, between the 12th and 13th values → $\dfrac{19+20}{2} = 19.5$ ✓
- **6th decile (P_{60}):** $L_{60} = 0.6(17) = 10.2$, between the 10th (16) and 11th (17) values → $\dfrac{16+17}{2} = 16.5$

> **About D_6:** the location 10.2 is much closer to position 10 than to 11, so the video suggests the true answer should be closer to 16. **Linear interpolation** makes this precise: $16 + 0.2(17 - 16) = 16.2$. With small n these estimates are rough, and they become more accurate as n grows.

Going backwards - which percentile is a given value?

$$\text{percentile of value } v = \frac{x + 0.5\,y}{n} \times 100$$

- x = number of values **less than** v
- y = number of times v appears
- n = total number of values

Using the same 16-number list:

- **Value 12:** 5 values are below it and 12 appears once.

  $$\frac{5 + 0.5(1)}{16} \times 100 = 34.375 \approx 34\text{th percentile}$$

  This makes sense: 12 lies between Q_1 (25th) and Q_2 (50th).
- **Value 16:** 8 values are below it and 16 appears twice.

$$\frac{8 + 0.5(2)}{16} \times 100 = 56.25 \approx 56\text{th percentile}$$

Percentiles from a cumulative relative frequency table

Data (20 values):

Value	f	RF	CRF
2	2	0.10	0.10
3	4	0.20	0.30
4	3	0.15	0.45
6	2	0.10	0.55
7	1	0.05	0.60
9	5	0.25	0.85
10	2	0.10	0.95
12	1	0.05	1.00
Total	**20**	**1.00**	

Rules:

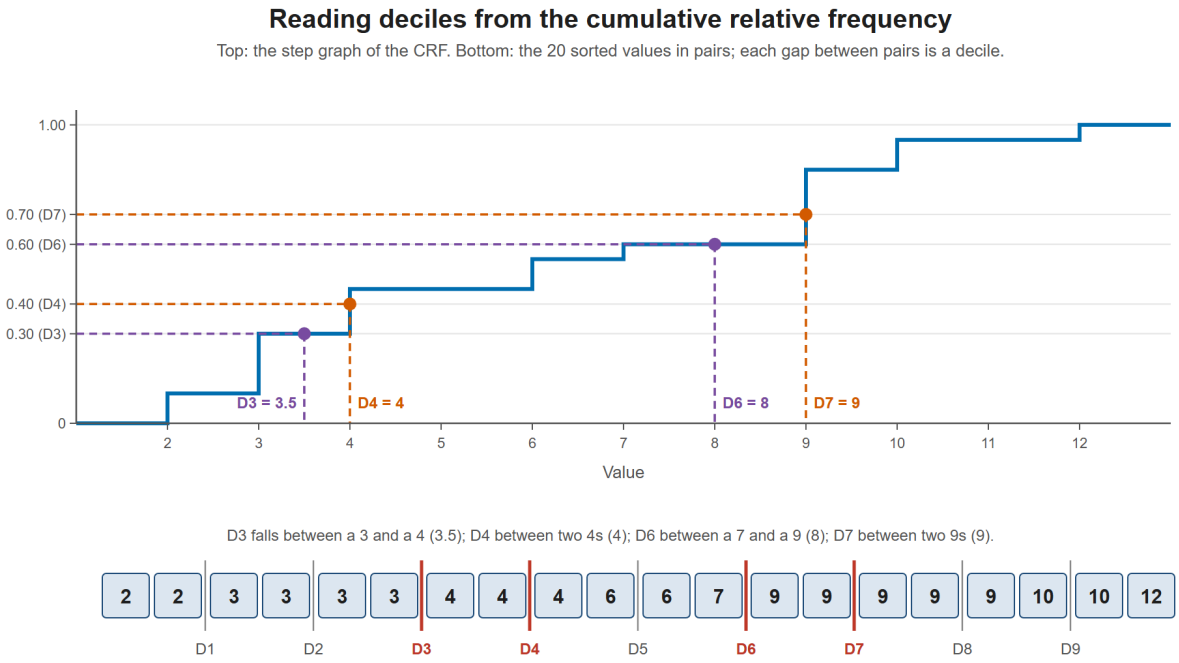
- If the percentile falls **strictly between** two CRF values, take the value of the **higher** row.
- If it equals a CRF value **exactly**, average that row's value with the next row's value.

Decile	Percentile	Where it falls	Value
D_4	40th	between 0.30 and 0.45	**4**
D_7	70th	between 0.60 and 0.85	**9**
D_3	30th	exactly 0.30 → average 3 and 4	**3.5**
D_6	60th	exactly 0.60 → average 7 and 9	**8**

Check by listing the data in pairs (20 values ÷ 10 deciles = 2 values per decile):

```
2 2 | 3 3 | 3 3 | 4 4 | 4 6 | 6 7 | 9 9 | 9 9 | 9 10 | 10 12
   D1    D2    D3    D4    D5    D6    D7    D8     D9
```

- D_3 sits between a 3 and a 4 → 3.5
- D_4 sits between two 4s → 4
- D_6 sits between a 7 and a 9 → 8
- D_7 sits between two 9s → 9



Reading deciles from the cumulative relative frequency. Where a decile level crosses a flat step, it lands strictly between two values, so take the higher one (D4 = 4, D7 = 9). Where it lands exactly on a step, average the two values on either side (D3 = 3.5, D6 = 8). The row of boxes below shows the same thing with the 20 sorted values.

Key takeaways

- Quartiles split the data into 4 parts, deciles into 10, percentiles into 100. $Q_2 = D_5 = P_{50}$ = median.
- Location of P_k: $\frac{k}{100}(n+1)$. If it is not a whole number, use the neighboring values (average them, or interpolate).
- Percentile of a value: $\frac{x + 0.5y}{n} \times 100$.
- From a CRF table: strictly between two rows → higher value; exactly on a row → average with the next value.
- Different textbooks and software use slightly different percentile rules. Small differences are normal.

Lesson 4.2 - How to Find the Interquartile Range

> **Video:** How To Find The Interquartile Range & any Outliers by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

Key ideas

```
min          Q1           Q2           Q3          max
 |-----------|------------|------------|-----------|
 0th        25th         50th         75th       100th percentile
             <------ middle 50% (IQR) ----->
```

- Q_2 = median of all the data (50th percentile).
- Q_1 = median of the **lower half** (25th percentile). Q_3 = median of the **upper half** (75th percentile).
- **Interquartile range:** $\text{IQR} = Q_3 - Q_1$. It is the spread of the **middle 50%** of the data.

Outlier test (1.5 × IQR rule)

$$\text{lower fence} = Q_1 - 1.5\,\text{IQR}, \qquad \text{upper fence} = Q_3 + 1.5\,\text{IQR}$$

Any data value **outside** $[\text{lower fence}, \text{upper fence}]$ is an **outlier**.

Example 1 - odd n, no outlier

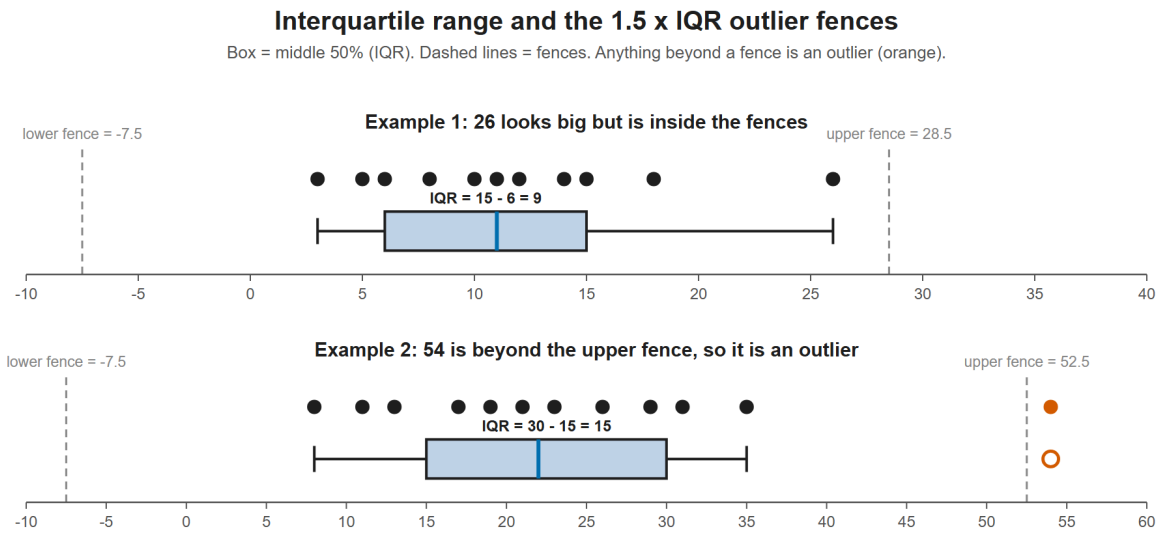
Data: 5, 8, 15, 26, 10, 18, 3, 12, 6, 14, 11

Sorted: 3, 5, **6**, 8, 10, **11**, 12, 14, **15**, 18, 26 ($n = 11$)

- $Q_2 = 11$ (the middle value, left out of both halves)
- Q_1 = median of 3, 5, 6, 8, 10 = **6**
- Q_3 = median of 12, 14, 15, 18, 26 = **15**
- $\text{IQR} = 15 - 6 = 9$

Fences: $6 - 1.5(9) = 6 - 13.5 = -7.5$ and $15 + 1.5(9) = 15 + 13.5 = 28.5$

26 looks large, but it is inside $[-7.5, 28.5]$, so it is **not an outlier**. A value of 29 would have been one.



The box spans the IQR (the middle 50% of the data) and the dashed fences sit 1.5 IQR beyond each end of it. In Example 1 every value, including 26, is inside the fences. In Example 2, 54 is past the upper fence of 52.5, so it is drawn as a separate outlier point.

Example 2 - even n, one outlier

Data: 11, 31, 21, 19, 8, 54, 35, 26, 23, 13, 29, 17

Sorted: 8, 11, 13 | 17, 19, 21 || 23, 26, 29 | 31, 35, 54 ($n = 12$)

- $Q_2 = \dfrac{21 + 23}{2} = 22$
- $Q_1 = \dfrac{13 + 17}{2} = 15$ (lower half: 8, 11, 13, 17, 19, 21)
- $Q_3 = \dfrac{29 + 31}{2} = 30$ (upper half: 23, 26, 29, 31, 35, 54)
- $\text{IQR} = 30 - 15 = 15$

Fences: $15 - 22.5 = -7.5$ and $30 + 22.5 = 52.5$

54 is above 52.5, so **54 is an outlier** ✓.

Key takeaways

- Sort the data → find Q_2 → find Q_1 and Q_3 from the two halves → IQR = $Q_3 - Q_1$.
- With even n (or an even-sized half), average the two middle values.
- Outlier fences: $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$. Check any value that "looks" extreme against them.

Lesson 4.3 - The Coefficient of Quartile Deviation

> **Video:** How To Calculate the Coefficient of Quartile Deviation by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

Formulas

Location of the k-th quartile in the sorted data:

$$\text{position of } Q_k = \frac{k(n+1)}{4}$$

The formula gives the **position**. Read the **value** from the data.

Quantity	Formula
Interquartile range	$\text{IQR} = Q_3 - Q_1$
Quartile deviation (semi-interquartile range)	$\text{QD} = \frac{Q_3 - Q_1}{2}$
Coefficient of quartile deviation	$\text{CQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Example

The data set has $n = 11$ values (already sorted). Its 3rd value is 14 and its 9th value is 35.

Find the quartiles.

- Q_1: position $\frac{1(11+1)}{4} = 3$ → 3rd value → $Q_1 = 14$
- Q_3: position $\frac{3(11+1)}{4} = 9$ → 9th value → $Q_3 = 35$

(a) Interquartile range

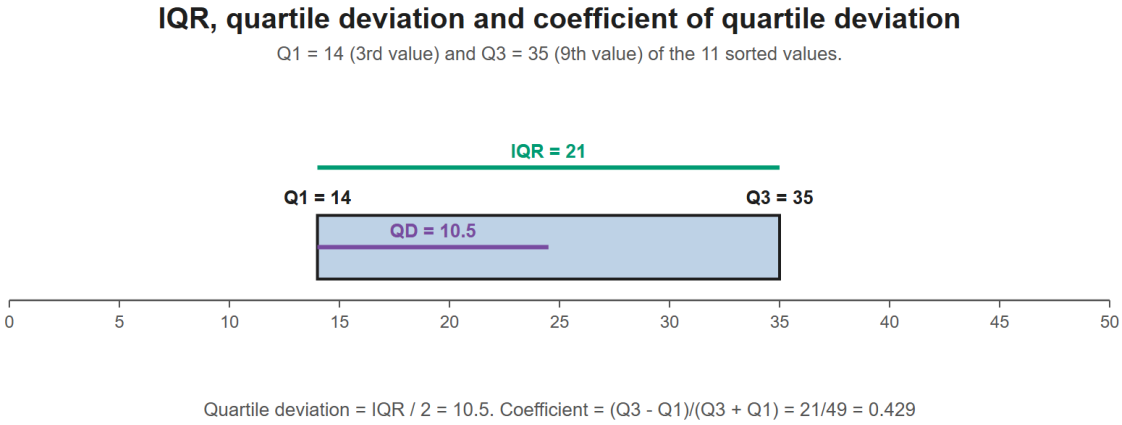
$$\text{IQR} = 35 - 14 = 21$$

(b) Quartile deviation

$$\text{QD} = \frac{21}{2} = 10.5$$

(c) Coefficient of quartile deviation

$$\text{CQD} = \frac{35 - 14}{35 + 14} = \frac{21}{49} \approx 0.429$$



The IQR is the width of the middle half of the data (35 - 14 = 21). The quartile deviation is half of that (10.5). The coefficient of quartile deviation divides the IQR by Q3 + Q1, which removes the units.

Key takeaways

- QD = half the IQR. It is a spread measure that ignores the extreme 25% at each end.
- CQD = $(Q_3 - Q_1)/(Q_3 + Q_1)$ has **no units**, so it can compare spread across data sets, like the coefficient of range ($(H - L)/(H + L)$, Lesson 3.4).

Lesson 4.4 - How to Make Box and Whisker Plots

> **Video:** How To Make Box and Whisker Plots by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

The five-number summary

A box-and-whisker plot (box plot) is built from five values:

1. **Minimum** (smallest value that is not an outlier)
2. Q_1
3. Q_2 (median)
4. Q_3
5. **Maximum** (largest value that is not an outlier)

Steps

1. Sort the data.
2. Find Q_2, then Q_1 and Q_3.
3. Compute $\text{IQR} = Q_3 - Q_1$ and the fences $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$.
4. Values outside the fences are **outliers**. They are plotted as **separate points**, not as part of the whiskers.
5. Draw a number line. Draw a **box** from Q_1 to Q_3 with a line at Q_2, and **whiskers** out to the minimum and maximum non-outliers.

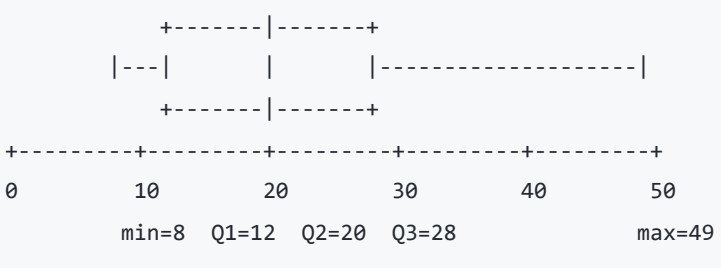
Example 1 - no outliers

Data: 11, 22, 20, 14, 29, 8, 35, 27, 13, 49, 10, 24, 17

Sorted ($n = 13$): 8, 10, 11, 13, 14, 17, **20**, 22, 24, 27, 29, 35, 49

- $Q_2 = 20$
- Lower half 8, 10, 11, 13, 14, 17 → $Q_1 = \dfrac{11 + 13}{2} = 12$
- Upper half 22, 24, 27, 29, 35, 49 → $Q_3 = \dfrac{27 + 29}{2} = 28$
- $\text{IQR} = 28 - 12 = 16$
- Fences: $12 - 1.5(16) = -12$ and $28 + 1.5(16) = 52$

Both 8 and 49 are inside $[-12, 52]$, so there are **no outliers**. Minimum = 8, maximum = 49.



Example 2 - with an outlier

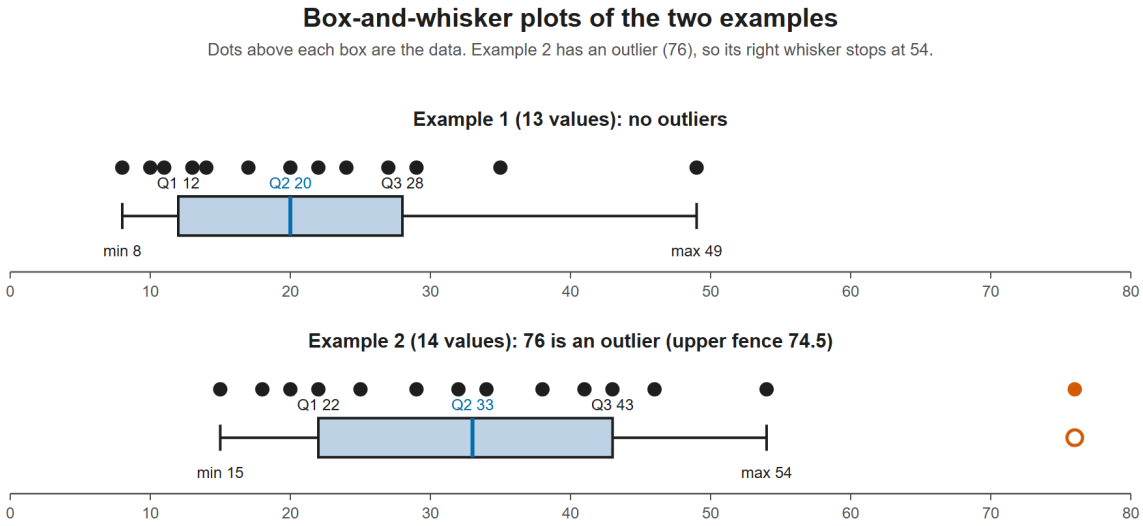
Data: 18, 34, 76, 29, 15, 41, 46, 25, 54, 38, 20, 32, 43, 22

Sorted ($n = 14$): 15, 18, 20, **22**, 25, 29, 32 || 34, 38, 41, **43**, 46, 54, 76

- $Q_2 = \dfrac{32 + 34}{2} = 33$
- Lower half (7 values) → middle value $Q_1 = 22$
- Upper half (7 values) → middle value $Q_3 = 43$
- $\text{IQR} = 43 - 22 = 21$
- Fences: $22 - 1.5(21) = -9.5$ and $43 + 1.5(21) = 74.5$

76 > 74.5, so 76 is an outlier. The right whisker stops at the largest non-outlier, **54**, and 76 is drawn as a separate point (`*`).

```
                    +----------|---------+
               |------|         |         |----------|                    *
                    +----------|---------+
+---------+---------+---------+---------+---------+---------+---------+---------+
0         10        20        30        40        50        60        70        80
             min=15  Q1=22    Q2=33    Q3=43     max=54              outlier=76
```



Both examples drawn to scale with the data above each box. In Example 2, 76 is beyond the upper fence of 74.5, so the right whisker stops at the largest non-outlier (54) and 76 is drawn as its own point.

Key takeaways

- Box = Q_1 to Q_3 (the IQR), with a line at the median Q_2.
- Whiskers reach the most extreme values that are **not** outliers.
- Outliers (beyond $Q_1 - 1.5\,\text{IQR}$ or $Q_3 + 1.5\,\text{IQR}$) are shown as individual points.

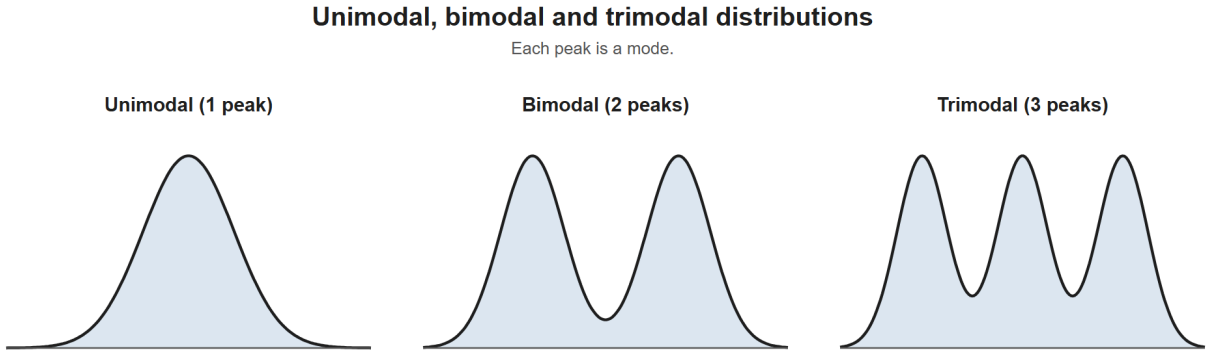
Lesson 4.5 - Skewness with Box Plots

> **Video:** Skewness - Right, Left & Symmetric Distribution - Mean, Median, & Mode With Boxplots by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

Number of modes

The **mode** is the value with the highest frequency, which is the **peak** of the distribution.

Shape	Peaks	Name
One peak	1	**Unimodal**
Two peaks	2	**Bimodal**
Three peaks	3	**Trimodal** (or, in general, **multimodal**)



The number of peaks gives the number of modes.

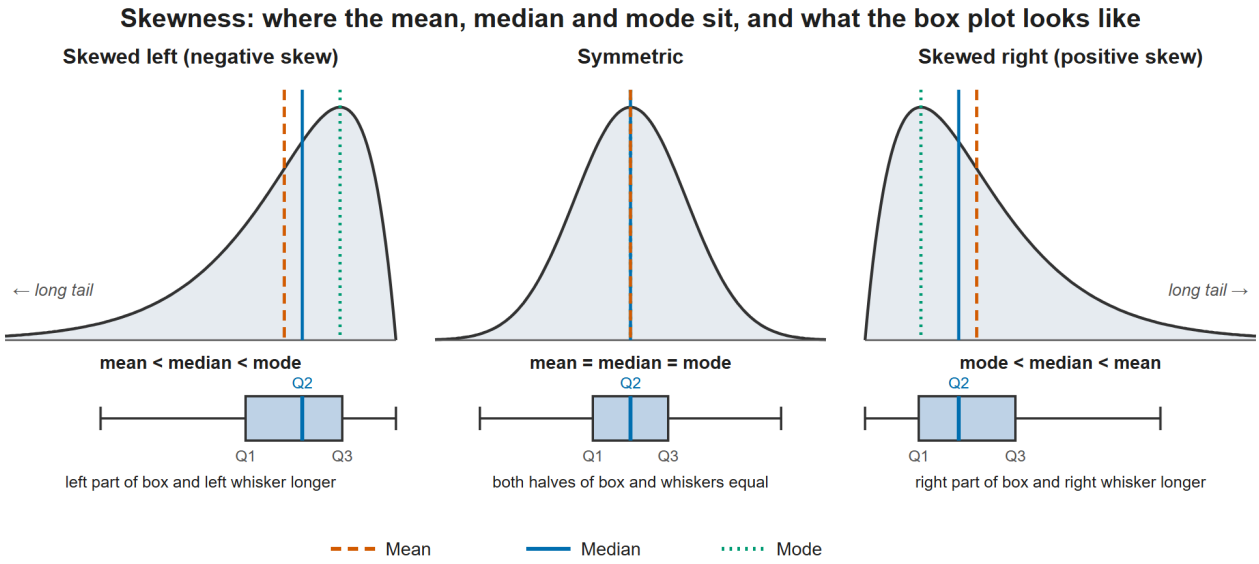
The three shapes (unimodal distributions)

Shape	Tail	Order of mean, median, mode
Symmetric	Left and right sides mirror each other	mean = median = mode
Skewed right (positive skew)	Long tail to the **right**	mode < median < **mean**
Skewed left (negative skew)	Long tail to the **left**	**mean** < median < mode

Reading left to right along the x-axis:

- **Skewed right:** peak (mode) → median → mean → long tail on the right
- **Skewed left:** long tail on the left → mean → median → peak (mode)

> **Memory aid:** the **mean is pulled toward the tail**. The mode stays at the peak, and the median sits between them.



Curves: Gamma(2, 1), a normal curve, and the mirror image of Gamma(2, 1). Whiskers drawn to the 1.5 × IQR fences.

Where the mean, median and mode fall for each shape, with the matching box plot underneath. The mean is always pulled toward the tail.

Reading skewness from a box plot

Let the left part of the box be $Q_2 - Q_1$ and the right part be $Q_3 - Q_2$.

Symmetric

```
|-------[=====|=====]-------|
```

- $Q_3 - Q_2 = Q_2 - Q_1$ (equal halves of the box)
- Whiskers of equal length

Skewed right (positive)

Any of these indicates right skew:

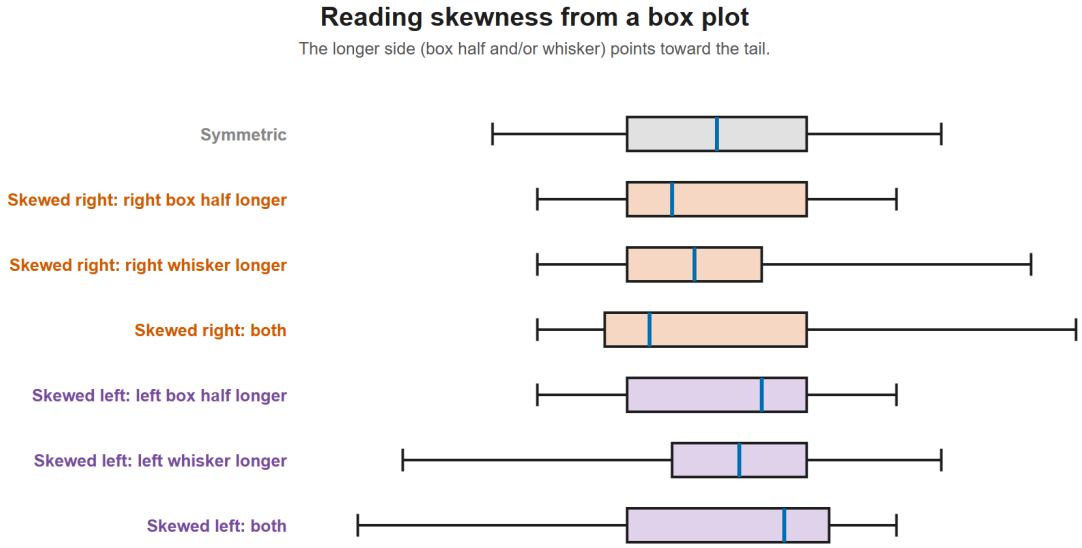
```
(a) right half of box longer:       |----[==|=======]----|
(b) equal box halves, longer right
    whisker:                        |--[====|====]----------|
(c) both:                           |--[==|========]------------|
```

- $Q_3 - Q_2 > Q_2 - Q_1$, and/or the right whisker is longer.

Skewed left (negative)

```
(a) left half of box longer:        |----[=======|==]----|
(b) equal box halves, longer left
    whisker:                        |----------[====|====]--|
(c) both:                           |------------[========|==]--|
```

- $Q_2 - Q_1 > Q_3 - Q_2$, and/or the left whisker is longer.



The three ways a box plot can show skew: a longer box half, a longer whisker, or both. The longer side always points toward the tail.

Key takeaways

- Unimodal, bimodal, multimodal = one, two, many peaks.
- Symmetric: mean = median = mode.
- Right skew: mean > median > mode. Left skew: mean < median < mode.
- On a box plot, the **longer side** (box half and/or whisker) points in the direction of the skew.